

# JACOB HOTZ

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CS student with hands-on range across robotics, systems, and full-stack development, and a fast track record of picking up new languages and tools on the job. Experienced leading and mentoring teams and explaining technical work clearly under real deadlines.

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## EDUCATION

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**Colorado School of Mines — Golden, CO — Expected Graduation: 2028**

B.S. in Computer Science, Focus in Robotic & Intelligent Systems | Minor in Electrical Engineering (Information & Systems Sciences)

## EXPERIENCE

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**FRC Team 4738, San Diego, CA — Lead Programmer**

- Trained and mentored a team of 10+ inexperienced programmers, teaching core programming concepts, robotics software design, and problem-solving strategies
- Guided students through development of competition-ready robot software via hands-on instruction, debugging sessions, and design reviews
- Led the programming team through two international competitions involving 3,300+ participating teams

**Alpha Tau Omega Fraternity, Golden, CO — Recruitment Chair**

- Recruited an average of ~20 new members per semester over the past two semesters for a 90-member chapter
- Managed high-stress situations, resolving emergent issues in real time and implementing solutions to maintain smooth operations
- Led a team of members to make decisions quickly and efficiently within short time frames

**Associated Students, Mission Bay, CA — Sailing Instructor**

- Delivered structured sailing instruction to children and adults while maintaining responsibility for high-value equipment
- Worked directly with clients in a service-oriented instructional role, adapting communication styles to varied skill levels

## PROJECTS

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**FRC Robot Code | Crescendo 2024 — Java**

- Developed competition-ready robot software for the 2024 FRC season using object-oriented design and swerve-drive kinematics for autonomous and driver-controlled modes
- Implemented odometry-based pose estimation and collaborated with hardware and controls teams while mentoring junior programmers on code structure and robot control logic

**2D Robotic Arm Simulation — Rust**

- Building a 2D robotic arm simulator with a WPILib-style architecture, focused on control systems and kinematic behavior
- Implemented a custom PID controller to regulate joint motion and system stability
- Designed the project to support interactive development and future feature expansion

**Sign-In Site — React, Firebase**

- Designed and implemented a full-stack authentication system with a React frontend and Firebase backend
- Integrated user authentication, data storage, and secure session handling

**Othello — Java, MVC**

- Built an Othello game using an MVC architecture to separate game logic, UI, and control flow
- Implemented a Minimax algorithm with Alpha-Beta pruning to optimize AI move selection

**Physics Engine — Java, Maven**

- Developed a basic physics engine implementing motion updates and force application
- Structured the project using Maven for dependency management and modular design

**FRC Robot Code | Charged Up 2023 — Java**

- Contributed to object-oriented robot control software under competition constraints during the 2023 season, implementing arm kinematics for game-piece placement and AprilTag-based vision alignment
- Assisted with debugging, testing, and performance tuning during build and competition phases

## SKILLS & STRENGTHS

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**Languages:** C, C++, Rust, Python, Java

**Web:** JavaScript, React, HTML, CSS

**Concepts:** Data Structures, Algorithms, Computer Organization, Embedded Systems

**Tools:** Git, Bash/Linux, Docker, VS Code, Maven, Firebase, WPILib